T.A.U. Engineering

	DIFFERENTIAL EQUATIONS Midterm I	
Course : *Differential Equations* Year : *2025-2026* Semester : *Fall* Lecturer : *Derya Altıntan* Date : *14 November 2025* Time : *15:00* Duration : *120 minutes*	Surname : Name : Number : Signature :	Department:
	7 QUESTION 100 POINTS	
1 | 2 | 3 | 4 | 5 | 6 | 7	TOTAL	

1.(20 pts)Obtain the particular solution of the following initial value problem

$$(y^2 - 3y + 2)\,dx - x\,dy = 0, \qquad y(1) = 3.$$

$(y^2-3y+2)\,dx - x\,dy = 0 \quad y(1)=3$ (Seperable)

$$\frac{dx}{x} = \frac{dy}{y^2-3y+2} = \frac{dy}{(y-1)(y-2)}$$

$$\frac{1}{y^2-3y+2} = \frac{A}{(y-2)} + \frac{B}{(y-1)}$$

$A=-1 \quad B=1$

$A(y-2)+B(y-1)=1$

$(A+B)y - 2A - B = 1$

$A+B=0 \qquad 2A+B=-1$

$A=-B \qquad -2B+B=-1$

$B=1 \qquad A=-1$

$$\frac{1}{y^2-3y+2} = -\frac{1}{y-1} + \frac{1}{y-2}$$

$$\int \frac{dx}{x} = \int \left(-\frac{1}{y-1} + \frac{1}{y-2}\right) dy$$

$$\ln|x| + \ln C = -\ln|y-1| + \ln|y-2|$$

$$C|x| = \frac{|y-2|}{|y-1|} \qquad y(1)=3 \quad C|1| = \frac{|3-2|}{|3-1|} \quad C = 1/2$$

$$\frac{1}{2}|x| = \frac{|y-2|}{|y-1|}$$

2. (20 pts) Obtain general solution of the following differential equation

$$y\,dx + (y^2 - x - 3)\,dy = 0.$$

$M_y = 1 \quad M_x = -1$ (Integrating factor)

$\frac{M_y - M_x}{-M} = \frac{1-(-1)}{-y} = \frac{-2}{y}$ $\quad \frac{d\mu}{\mu} = -\frac{2}{y}dy$

$\ln \mu(y) = -2\ln y \qquad \mu(y) = \frac{1}{y^2}$

$\frac{1}{y^2}\cdot y\,dx + \frac{1}{y^2}(y^2 - x - 3)\,dy = 0$

$\frac{1}{y}\,dx + \left(1 - \frac{x}{y^2} - \frac{3}{y^2}\right)dy = 0$

$A = \frac{1}{y}$ $\quad A_y = -\frac{1}{y^2}$ $\quad B = 1 - \frac{x}{y^2} - \frac{3}{y^2}$ $\quad B_x = -\frac{1}{y^2}$ Exact df

$F(x,y) = \int \frac{1}{y}\,dx + h(y) \qquad F(x,y) = \frac{x}{y} + h(y)$

$\frac{\partial F}{\partial y} = -\frac{x}{y^2} + h'(y) = 1 - \frac{x}{y^2} - \frac{3}{y^2}$

$h'(y) = y + \frac{3}{y}$

$F(x,y) = \frac{x}{y} + \frac{y^2}{2} + 3\ln y$

3.(5+5 pts) For the following Initial Value Problems (IVPs), determine the region or interval where the conditions of the existence and uniqueness theorem are satisfied.

(a)

$$(t^2 - 9)y' + y = \frac{t}{\ln(t)}, \quad y(4) = 1$$

$y' + \frac{1}{t^2-9}y = \frac{t}{\ln t(t^2-9)}$ $\quad p(t)$ $\quad q(t)$

Since we have $y(4) = 1$. The largest interval containing 4 is $(3, \infty)$

$y \in (-\infty, \infty)$

$R = (3,\infty) \times (-\infty, \infty)$

$p(t)$: discont at $t = \mp 3$, cont at $t \in \mathbb{R} \setminus \{-3, 3\}$

$q(t) = \frac{t}{\ln t(t^2-9)}$ discont at $t = 1$, $t = \mp 3$ but since $\ln t$ is only defined for $t > 0$.

$q(t)$ is cont at $t \in (0,1) \cup (1,3) \cup (3,\infty)$

(b)

$$y' = \frac{\sqrt{y-1}}{t^2 - 16}, \quad y(0) = 2$$

$f(t,y) = \frac{\sqrt{y-1}}{t^2-16}$ cont for $t \in \mathbb{R} \setminus \{-4, 4\}$

$f_y(t,y) = \frac{1}{2\sqrt{y-1}(t^2-16)}$ $\quad y \neq 1$ $\quad \mathbb{R} \setminus \{1\}$

Then $R = (-4, 4) \times (1, \infty)$

$0 \in$ $\quad 2 \in$

4.(20 pts) Find the general solution $x(t)$ to the homogeneous linear differential equation system $x' = Ax$ defined by the matrix:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 2 & 1 & 5 \end{pmatrix}$$

Eigenvalues $\det(A-\lambda I) = |A-\lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{vmatrix}$

$= (1-\lambda)(-1)^{1+1} \begin{vmatrix} -3-\lambda & 0 \\ 1 & 5-\lambda \end{vmatrix} = \underbrace{(1-\lambda)}_{\lambda=1}\underbrace{(-3-\lambda)}_{\lambda=-3}\underbrace{(5-\lambda)}_{\lambda=5}$

$\lambda_1 = 1$ $A\vec{v} = 1\cdot\vec{v}$

$(A-I)\vec{v} = 0$ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & -4 & 0 \\ 2 & 1 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 4 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 2 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-1/4 R_2 \to R_2} \begin{bmatrix} 1 & 1/2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ leading 1

$v_1 + \frac{1}{2}v_2 + 2v_3 = 0$

$v_2 = 0$

$v_1 + 2v_3 = 0 \Rightarrow v_1 = -2v_3$

v_1, v_2: leading

v_3: free

$\vec{v} = \begin{bmatrix} -2v_3 \\ 0 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$

$V_{\lambda_1=1} = \text{span}\left\{\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}\right\}$

$\lambda_2 = -3$ $A\vec{v} = -3\vec{v}$

$(A+3I)\vec{v} = 0$ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & 1 & 8 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 8 \\ 0 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}$

$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 8 \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 4 \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 - 4R_1 \to R_2} \begin{bmatrix} 1 & 1/2 & 4 \\ 0 & -2 & -16 \\ 0 & 0 & 0 \end{bmatrix}$

$\xrightarrow{-1/2 R_2 \to R_2} \begin{bmatrix} 1 & 1/2 & 4 \\ 0 & 1 & 8 \\ 0 & 0 & 0 \end{bmatrix}$

v_1, v_2: leading

v_3: free

$v_1 + \frac{v_2}{2} + 4v_3 = 0$

$v_2 + 8v_3 = 0$

$v_1 = 0$

$\Rightarrow v_2 = -8v_3$

$V = v_3 \begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix}$

$V_{\lambda=-3} = \text{span}\left\{\begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix}\right\}$

$\lambda = 5$ $(A-5I)\vec{v} = ?$

$\begin{bmatrix} -4 & 0 & 0 \\ 0 & -8 & 0 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 0 \\ 0 & -8 & 0 \\ -4 & 0 & 0 \end{bmatrix}$

$\xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & -8 & 0 \\ -4 & 0 & 0 \end{bmatrix} \xrightarrow{4R_1 + R_3 \to R_3} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & -8 & 0 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow{R_2/2 \to R_2} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$

$\xrightarrow{-2R_2 + R_3 \to R_3} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

v_1, v_2: leading

v_3: free

$v_1 + \frac{v_2}{2} = 0$

$v_2 = 0$

$v_1 = 0$ $v_3 = 1$

$V_{\lambda=5} = \text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$

$x_{\text{general}} = c_1 e^{t}\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{-3t}\begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix} + c_3 e^{5t}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

5. (10+10 pts) Consider the following linear system and initial condition:

$$\mathbf{x}' = \begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix}\mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

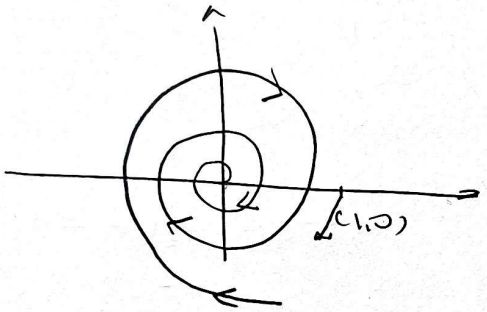
(a) Find the eigenvalues of matrix A, draw phase portrait, determine the type and stability of the equilibrium point Node).

(b) Solve the Initial Value Problem (IVP) to find the specific solution $\mathbf{x}(t)$.

(a) $\det(A-\lambda I) = \begin{vmatrix} -3-\lambda & 2 \\ -2 & -3-\lambda \end{vmatrix} = (-3-\lambda)^2 - 2(-2) = 0$

$= 9+6\lambda+\lambda^2+4 = \lambda^2+6\lambda+13$

$\lambda^2+6\lambda+13 = 0 \quad \lambda_{1,2} = \dfrac{-6 \mp \sqrt{6^2-4\cdot1\cdot13}}{2} = \dfrac{-6 \mp \sqrt{36-52}}{2} = -3 \mp 2i$



stable spiral $\quad Re(\lambda) = -3 < 0.$

$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ -2 & -3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$

(b) $\lambda_1 = -3+2i \quad (A-\lambda_1 I)\vec{v} \quad (A-(-3+2i)I)\vec{v} = ?$

$\begin{pmatrix} -3-(-3+2i) & 2 \\ -2 & -3-(-3+2i) \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$-2iv_1 + 2v_2 = 0$

$-2v_1 - 2iv_2 = 0 \Rightarrow v_1 = -iv_2$

$v_2 = i \quad v_1 = 1$

$V_{\lambda=(-3+2i)} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ i \end{bmatrix}\right\}$

$\vec{z}(t) = e^{(-3+2i)t}\begin{bmatrix} 1 \\ i \end{bmatrix} = e^{-3t}e^{2it}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + i\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$

$= e^{-3t}(\cos 2t + i\sin 2t)\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + i\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = e^{-3t}\left(\begin{bmatrix} \cos 2t \\ 0 \end{bmatrix} + i\begin{bmatrix} 0 \\ \cos 2t \end{bmatrix}\right.$

$\left. + i\begin{bmatrix} \sin 2t \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \sin 2t \end{bmatrix}\right) = e^{-3t}\left(\underbrace{\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix}}_{x_1} + i\underbrace{\begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix}}_{x_2}\right)$

$x_p(t) = c_1 e^{-3t}\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix} + c_2 e^{-3t}\begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix}$

$\boxed{x(t) = e^{-3t}\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix}} \quad c_1 = 1$

$x(0) = c_1 e^0\begin{bmatrix} \cos 0 \\ -\sin 0 \end{bmatrix} + c_2 e^0\begin{bmatrix} \sin 0 \\ \cos 0 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad c_2 = 0$

(5×1pts) The parts below give eigenvalues for a system of equations. Match each set of eigenvalues with a possible phase portrait (write one phase portrait letter in each box).

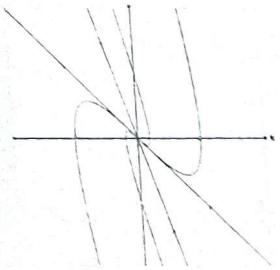
☐ $\lambda_1 = -1,\ \lambda_2 = 2$ (B)

☐ $\lambda_1 = -3,\ \lambda_2 = -1$ (A)

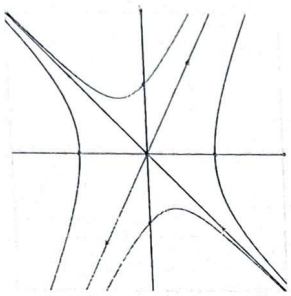
☐ $\lambda_1 = -i,\ \lambda_2 = i$ (E)

☐ $\lambda_1 = 1,\ \lambda_2 = 2$ (D)

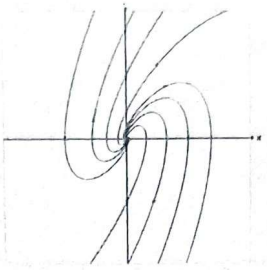
☐ $\lambda_1 = 1 - i,\ \lambda_2 = 1 + i$ (C)



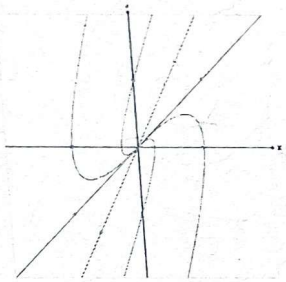
(a)



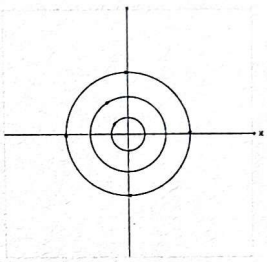
(b)



(c)



(d)



(e)

6. (5×1pts) The parts below give eigenvalues for a system of equations. Match each set of eigenvalues with a possible phase portrait (write one phase portrait letter in each box).

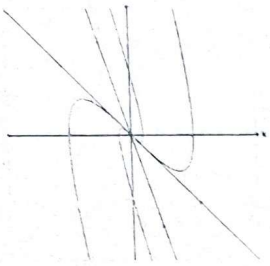
☐ $\lambda_1 = -1,\ \lambda_2 = 2$ B

☐ $\lambda_1 = -3,\ \lambda_2 = -1$ A

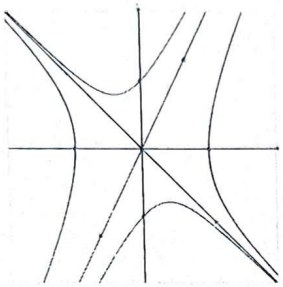
☐ $\lambda_1 = -i,\ \lambda_2 = i$ E

☐ $\lambda_1 = 1,\ \lambda_2 = 2$ D

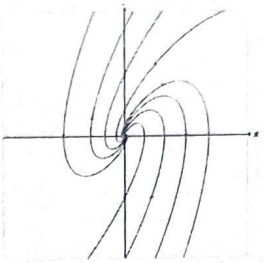
☐ $\lambda_1 = 1 - i,\ \lambda_2 = 1 + i$ C



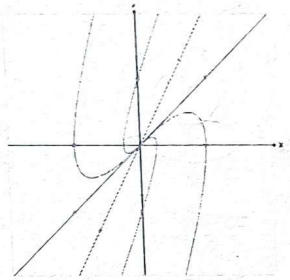
(a)



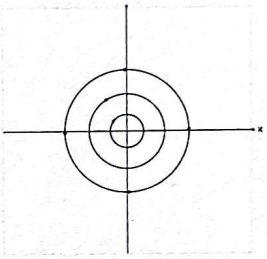
(b)



(c)



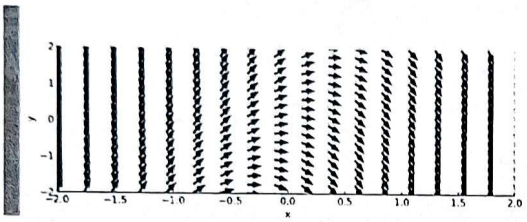
(d)



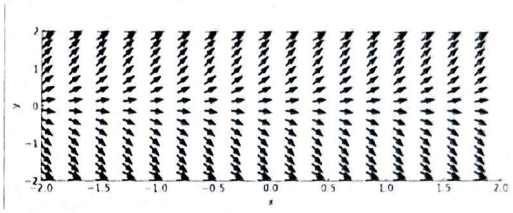
(e)

7. (5×1pts) . Match the following direction fields with their differential equations.

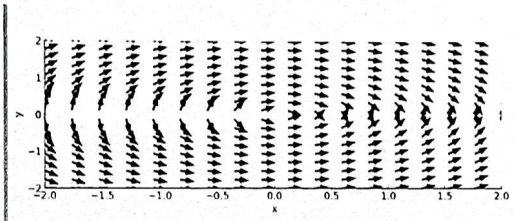
- [ ] $y' = x + 2y$ (D)
- [ ] $y' = -x/y$ (C)
- [ ] $y' = y - 5x$ (A)
- [ ] $y' = y(y-2)(y-1)$ (B)
- [ ] $y' = y/2x$ (E)



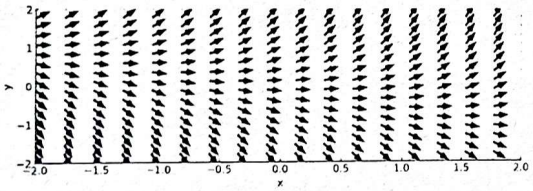
(a)



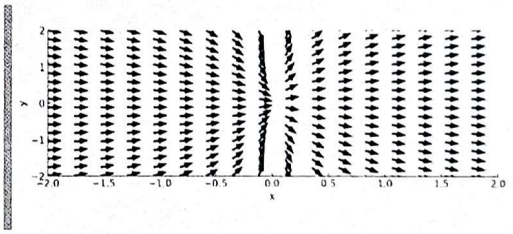
(b)



(c)



(d)



(e)

DERYA ALTINTAN

♡ GOOD LUCK ♡

TAU MATH 219 DIF. EQUATIONS MIDTERM I

(1) Obtain the particular solution of the following initial value problem

$$(y^2-3y+2)dx - x\,dy = 0 \qquad y(1)=3$$ (Homogeneous)

$$\frac{dx}{x} = \frac{dy}{y^2-3y+2} = \frac{dy}{(y-1)(y-2)}$$

$$\frac{1}{y^2-3y+2} = \frac{A}{y-1} + \frac{B}{y-2}$$

$$\frac{1}{(y-2)(y-1)}$$

$A(y-2)+B(y-1)=1$

$(A+B)y - 2A - B = 1$

$A+B=0 \qquad 2A+B=-1$

$A=-B \qquad -B=-1$

$B=1$

$A=-1$

$$\frac{1}{y^2-3y+2} = -\frac{1}{y-1} + \frac{1}{y-2}$$

$$\int \frac{dx}{x} = \int \left(-\frac{1}{y-1} + \frac{1}{y-2}\right) dy$$

$$\ln|x| + \ln C = -\ln|y-1| + \ln|y-2|$$

$$C|x| = \frac{|y-2|}{|y-1|} \qquad y(1)=3.$$

$$C.1 = \frac{|3-2|}{|3-1|} \qquad C = 1/2$$

$$\frac{1}{2}|x| = \frac{|y-2|}{|y-1|}$$

(2) Obtain general solution of the following differential equation $y dx + (y^2 - x - 3) dy = 0$ (Integrating factor)

$M_y = 1 \quad N_x = -1$

$$\frac{M_y - N_x}{-M} = \frac{1-(-1)}{-y} = -\frac{2}{y} \qquad \frac{d\mu}{\mu} = -\frac{2}{y} dy$$

$$\ln \mu(y) = -2 \ln y \qquad \mu(y) = \frac{1}{y^2}$$

$$\frac{1}{y^2} y\, dx + \frac{1}{y^2} (y^2 - x - 3) dy = 0$$

$$\frac{1}{y} dx + \left(1 - \frac{x}{y^2} - \frac{3}{y^2}\right) dy = 0$$

$A_y = -1/y^2 = B_x = -1/y^2$ Exact dif.

$$F(x,y) = \int \frac{1}{y} dx + h(y)$$

$$F(x,y) = \frac{x}{y} + h(y) \qquad \frac{\partial F}{\partial y} = -\frac{x}{y^2} + h'(y) = 1 - \frac{x}{y^2} - \frac{3}{y^2}$$

$$h'(y) = y + \frac{3}{y}$$

$$\boxed{F(x,y) = \frac{x}{y} + y + \frac{3}{y} = C}$$

(3) Find the general solution to the differential equation system $\dot{x} = Ax$ defined by

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 2 & 1 & 5 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{vmatrix} = (1-\lambda)(-1)^{1+1} \begin{vmatrix} -3-\lambda & 0 \\ 1 & 5-\lambda \end{vmatrix}$$

$$= (1-\lambda)(-3-\lambda)(5-\lambda)$$

$\lambda_1 = 1$: $A\vec{v} = 1\cdot\vec{v}$

$(A - I)\vec{v} = 0$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -4 & 0 \\ 2 & 1 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 4 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 2 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-1/4 R_2 \to R_2} \begin{bmatrix} 1 & 1/2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

leading 1

$v_1 + 1/2 v_2 + 2v_3 = 0$

$v_2 = 0$

$v_1 + 2v_3 = 0 \quad v_1 = -2v_3$

v_1, v_2: leading

v_3 : free.

$$\vec{v} = \begin{bmatrix} -2v_3 \\ 0 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$V_{\lambda=1} = \left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$\lambda_2 = -3$ $\quad A\vec{v} = -3\vec{v}$

$(A + 3I)\vec{v} = 0$

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & 1 & 8 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 8 \\ 0 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_2} \begin{bmatrix} 2 & 1 & 8 \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 4 \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 - 4R_1 \to R_2} \begin{bmatrix} 1 & 1/2 & 4 \\ 0 & -2 & -16 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-1/2 R_2 \to R_2} \begin{bmatrix} 1 & 1/2 & 4 \\ 0 & 1 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1, v_2 → leading

v_3 → free

$v_1 + 1/2 v_2 + 4v_3 = 0$

$v_2 + 8v_3 = 0$

$v_1 = 0.$

$\Rightarrow v_2 = -8v_3$

$$v = v_3 \begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix}$$

$$V_{\lambda=-3} = \text{span}\left\{ \begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix} \right\}$$

$\lambda=5$ $(A-5I)\vec{v}=0$

$$\begin{bmatrix} -4 & 0 & 0 \\ 0 & -8 & 0 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 0 \\ 0 & -8 & 0 \\ -4 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1/2} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & -8 & 0 \\ -4 & 0 & 0 \end{bmatrix} \xrightarrow{4R_1+R_3 \to R_3} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & -8 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 \to R_2/8} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow{-2R_2+R_3 \to R_3} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

v_1, v_2: leading $\quad v_1 + v_2/2 = 0 \quad v_1 = 0 \quad v_3 = 1$

v_3: free $\quad v_2 = 0$

$$V_{\lambda=5} = \text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$$

$$x_{general} = c_1 e^{t}\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{-3t}\begin{bmatrix} 0 \\ -8 \\ 1 \end{bmatrix} + c_3 e^{5t}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

(4) Solve the following initial value problem and determine the phase portrait for equilibrium solution.

$$x' = Ax \quad \text{where} \quad A = \begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix} \quad x(0) = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

$$\det(A-\lambda I) = \begin{vmatrix} -3-\lambda & 2 \\ -2 & -3-\lambda \end{vmatrix} = (-3-\lambda)^2 - (2)(-2) = 0$$

$$= 9 + 6\lambda + \lambda^2 + 4 = \lambda^2 + 6\lambda + 13$$

$$\lambda^2 + 6\lambda + 13 = 0$$

$$\lambda_{1,2} = \frac{-6 \mp \sqrt{6^2 - 4 \cdot 1 \cdot 13}}{2} = \frac{-6 \mp \sqrt{36-52}}{2} = -3 \mp 2i$$

$(A-(-3+2i)I)\vec{v}=0$

$$\begin{pmatrix} -3-(-3+2i) & 2 \\ -2 & -3-(-3+2i) \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$-2iv_1+2v_2=0$

$-2v_1-2iv_2=0 \Rightarrow v_1=-iv_2$

$v_2=i,\ v_1=1$

$$V_{\lambda=(-3+2i)}=\text{span}\left\{\begin{bmatrix} 1 \\ i \end{bmatrix}\right\}$$

$$z(t)=e^{(-3+2i)t}\left(\begin{bmatrix} 1 \\ i \end{bmatrix}\right)=e^{-3t}\cdot e^{2it}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}+i\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$=e^{-3t}\left((\cos 2t+i\sin 2t)\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}+i\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)\right)$$

$$=e^{-3t}\left(\begin{bmatrix} \cos 2t \\ 0 \end{bmatrix}+i\begin{bmatrix} 0 \\ \cos 2t \end{bmatrix}+i\begin{bmatrix} \sin 2t \\ 0 \end{bmatrix}-\begin{bmatrix} 0 \\ \sin 2t \end{bmatrix}\right)$$

$$=e^{-3t}\left(\underbrace{\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix}}_{x_1}+i\underbrace{\begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix}}_{x_2}\right)$$

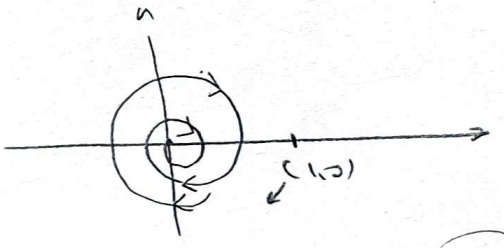
$$x_p(t)=c_1e^{-3t}\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix}+c_2e^{-3t}\begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix}$$

$$x(0)=c_1\cdot e^0\begin{bmatrix} \cos 0 \\ -\sin 0 \end{bmatrix}+c_2e^0\begin{bmatrix} \sin 0 \\ \cos 0 \end{bmatrix}=\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \end{bmatrix}\quad c_1=1,\ c_2=0$$

$$x(t)=e^{-3t}\begin{bmatrix} \cos 2t \\ -\sin 2t \end{bmatrix}$$

For $x_p(t)$ $t\to\infty$ $x_p(t)\to 0$ since $\text{Re}(\lambda)=-3<0$ spiral toward zero give a point $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$A\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix}=\begin{bmatrix} -3 & 2 \\ -2 & -3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}=\begin{bmatrix} -3 \\ -2 \end{bmatrix}$$



"Stable spiral"

(4) For the following questions find the largest region to guarantee existence and unique solution.

(a) $y'(t^2-9)y' + y = \frac{t}{\ln t}$ $\quad y(4)=1$.

$y' + \underbrace{\frac{1}{t^2-9}}_{p(t)}\, y = \underbrace{\frac{t}{\ln t \cdot (t^2-9)}}_{q(t)}$

$p(t)$: discont at $t = \pm 3$ $\quad p(t)$ cont $t \in \mathbb{R} \setminus \{-3, 3\}$

$q(t) = \frac{t}{\ln t\,(t^2-9)}$ discont at $t=1$ $t=\pm 3$ but since $\ln t$ is only defined for $t>0$

so $q(t)$ is cont $t \in (0,1) \cup (1,3) \cup (3,\infty)$

since we have initial point $y(4)=1$ The largest interval containing 4 is $(3,\infty)$ $\quad y \in (-\infty, \infty)$ $\quad \mathbb{R} = (3,\infty) \times (-\infty,\infty)$

(b) $y' = \frac{\sqrt{y-1}}{t^2-16}$ $\quad y(0)=2$.

$f(t,y) = \frac{\sqrt{y-1}}{t^2-16}$ cont for $\mathbb{R} \setminus \{-4, 4\}$.

$f_y(t,y) = \frac{1}{2\sqrt{y-1}\,(t^2-16)}$ $\quad y \neq 1$ $\quad \mathbb{R} \setminus \{1\}$.

Then $\mathcal{R} = \underbrace{(-4, 4)}_{0 \in} \times \underbrace{(1, \infty)}_{2 \in}$

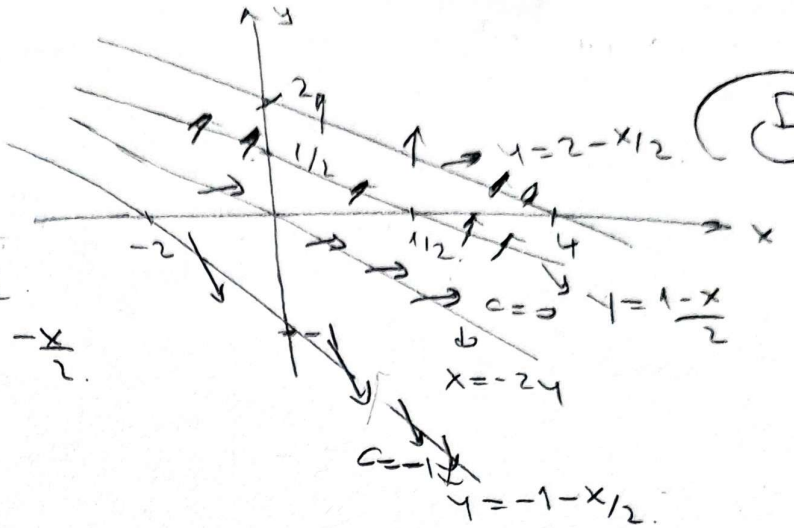
(6) $y' = x + 2y$

$x + 2y = 0 \quad x = -2y$

$x + 2y = 1 \Rightarrow y = \frac{1-x}{2}$

$x + 2y = 4 \Rightarrow y = 2 - \frac{x}{2}$

$x + 2y = -1 \Rightarrow y = -1 - \frac{x}{2}$



(D)

$y' = y(y-2)(y+2)$

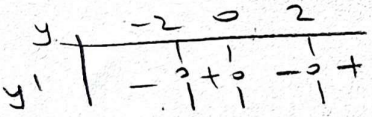
$y'=0 \quad y=0 \quad y=2 \quad y=-2$

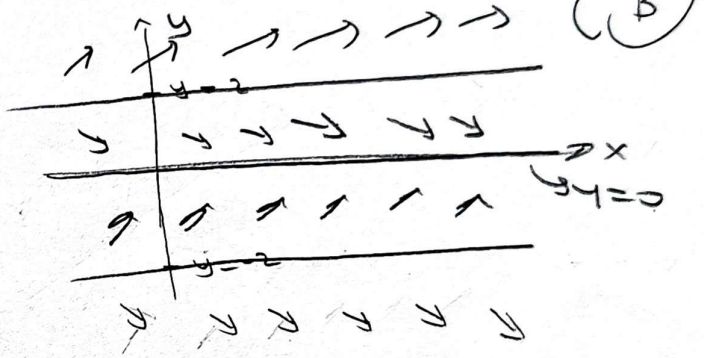
$0 < y < 2 \quad y' < 0$

$y > 2 \quad y' > 0$

$-2 < y < 0 \quad y' > 0$

$y < -2 \quad y' < 0$





(B)

$y' = \frac{-x}{y}$

$y'=0 \quad x=0$

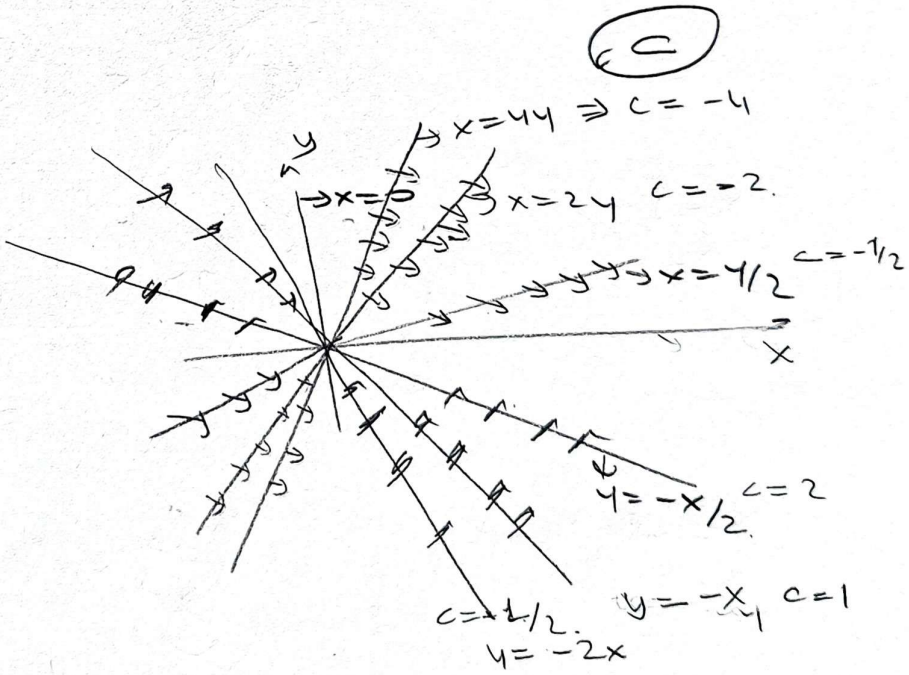
$y'=1 \quad -x=y$

$y'=2 \quad -x=2y$

$y'=-2 \quad x=2y$

$y'=-1 \quad x=y$

$y'=-1/2 \quad x=y/2$



(C)

$y' = \frac{y}{2x}$

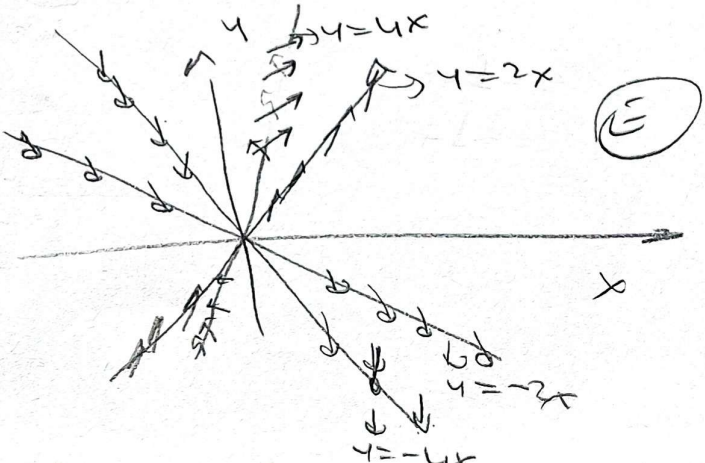
$y'=0 \quad y=0$

$y'=1 \quad y=2x$

$y'=2 \quad y=4x$

$y'=-1 \quad y=-2x$

$y'=-2 \quad y=-4x$



(E)

$=5x$

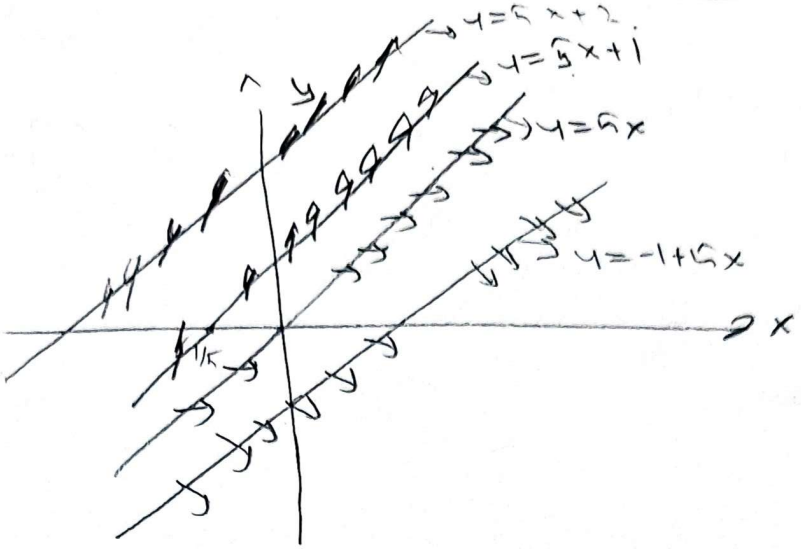
$y=1+5x$

$y=.2+5x$

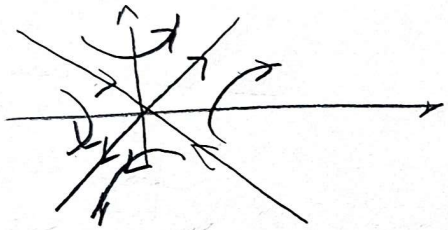
$y=-1+5x$

$y=-2+5x$

(A)

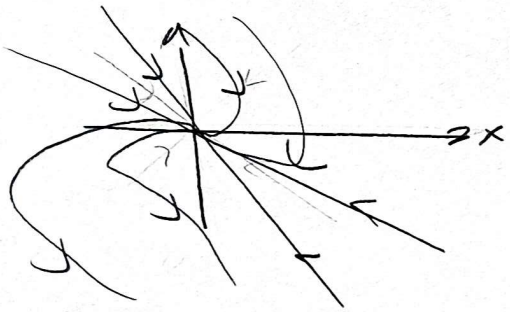


$=-1$
$\lambda=2$



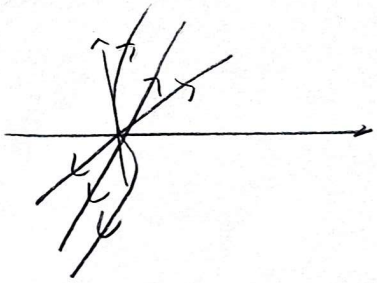
B = -1, 2

$\lambda=-3$
$\lambda=-1$



A = -3, -1

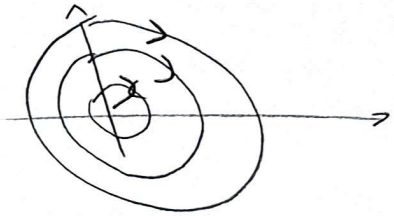
$\lambda=1$
$\lambda=2$



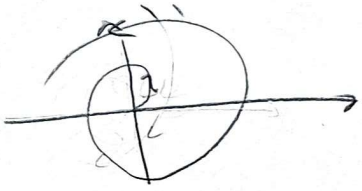
D = 1, 2

E = 7i

$\lambda=-i$
$\lambda=i$



$\lambda=-1-i$
$\lambda=-1+i$



C = -1 - i
-1 + i